form, however. If $A = SJS^{-1}$ is the Jordan canonical form of A, then $A^m = SJ^mS^{-1}$, so $A^m \to 0$ as $m \to \infty$ if and only if $J^m \to 0$ as $m \to \infty$. Since J is a direct sum of Jordan blocks, it suffices to consider the behavior of powers of a Jordan block

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix} = \begin{bmatrix} \lambda & & 0 \\ & \ddots & \\ 0 & & \lambda \end{bmatrix} + \begin{bmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{bmatrix}$$

$$= \lambda I + N_k \in M_k, \quad \text{where } N_k \equiv J_k(0)$$

Since $N_k^m = 0$ for all $m \geq k$, we have

$$[J_k(\lambda)]^m = (\lambda I + N_k)^m = \sum_{i=0}^{m} \binom{m}{i} \lambda^i N_k^{m-i} = \sum_{i=m-k+1}^{m} \binom{m}{i} \lambda^i N_k^{m-i}$$

for all $m \geq k$. Since the diagonal elements are all λ^m, if $J^m \to 0$ it is necessary that $\lambda^m \to 0$, which means that $|\lambda| < 1$. Conversely, if $|\lambda| < 1$, we would like to prove that

$$\binom{m}{m-j} \lambda^{m-j} \to 0 \quad \text{as} \quad m \to \infty \quad \text{for each} \quad j = 0, 1, 2, \ldots, k-1$$

But

$$\left| \binom{m}{m-j} \lambda^{m-j} \right| = \left| \frac{m(m-1)(m-2)\cdots(m-j+1)\lambda^m}{j!\lambda^j} \right| \leq \left| \frac{m^j \lambda^m}{j!\lambda^j} \right|$$

so it will suffice to show that $m^j |\lambda|^m \to 0$ as $m \to \infty$. An easy way to see this is to take logarithms and observe that

$$j \log m + m \log |\lambda| \to -\infty$$

as $m \to \infty$ because $\log|\lambda| < 0$ and $(\log x)/x \to 0$ as $x \to \infty$ by l'Hopital's rule.

This argument has made essential use of the Jordan canonical form of A to show that $A^m \to 0$ as $m \to \infty$ if and only if all the eigenvalues of A have modulus less than 1. Another proof, which is completely independent of the Jordan canonical form, is given in (5.6.12).

3.2.6 **The geometric multiplicity–algebraic multiplicity inequality.** The geometric multiplicity of an eigenvalue of a given matrix $A \in M_n$ is the number of Jordan blocks of A corresponding to the eigenvalue. This number is less than or equal to the sum of the orders of all the Jordan blocks corresponding to the eigenvalue. This sum is the algebraic multiplicity. Thus, the geometric multiplicity of an eigenvalue is not greater than its algebraic multiplicity; compare with (1.4.9).

3.2.7 **Diagonalizable and nilpotent matrices.** A matrix $A \in M_n$ is *nilpotent* if $A^k = 0$ for some positive integer k. Any Jordan block $J_k(\lambda)$ can be written as $J_k(\lambda) = \lambda I + N_k$, where $(N_k)^k = 0$. Thus, any Jordan block is the sum of a diagonal matrix and a nilpotent matrix.

More generally, a Jordan matrix (3.2.1.1) can be written as $J = D + N$, where D is a diagonal matrix whose main diagonal is the same as that of J, and $N = J - D$. The matrix N is nilpotent, and $N^k = 0$ if k is the order of the largest Jordan block in J.

Finally, if $A \in M_n$ is any given matrix and $A = SJS^{-1}$ is in Jordan canonical form, then $A = SDS^{-1} + SNS^{-1} \equiv A_D + A_N$, where A_D is diagonalizable, A_N is nilpotent, and $A_D A_N = A_N A_D$ because both D and N are block diagonal matrices with respective blocks of the same size, and the blocks in D are scalar matrices.

We conclude that any $A \in M_n$ can be written as the sum of a diagonalizable matrix and a nilpotent matrix in such a way that these summands commute.

Problems

1. Let $\mathcal{F} = \{A_\alpha : \alpha \in \mathcal{I}\} \subset M_n$ be a given family of matrices, indexed by the index set $\mathcal{I}$, and suppose there is a nonderogatory matrix $A_0 \in \mathcal{F}$ such that $A_\alpha A_0 = A_0 A_\alpha$ for all $\alpha \in \mathcal{I}$. Show that for every $\alpha \in \mathcal{I}$ there is a polynomial $p_\alpha(t)$ of degree at most $n-1$ such that $A_\alpha = p_\alpha(A_0)$, and hence $\mathcal{F}$ is a commuting family.

2. Let $A \in M_n$ be given, and let λ_i be an eigenvalue of A. Show that the order of the largest Jordan block of A corresponding to the eigenvalue λ_i (the *index* of λ_i) is the smallest value of $k = 1, 2, \ldots, n-1$ for which $\operatorname{rank}(A - \lambda_i I)^k = \operatorname{rank}(A - \lambda_i I)^{k+1}$.

3. If $A \in M_n$ has $A^k = 0$ for some $k > n$, show that $A^r = 0$ for some $r \leq n$. Thus, every nilpotent matrix has a vanishing power that is not greater than the order of the matrix. *Hint:* Show that 0 is the only eigenvalue of A. What does the Jordan canonical form of A look like? Now take powers.

4. Let $J_k(0)$ be a given Jordan block. Use the argument at the end of (3.2.1) to determine the three possible Jordan canonical forms of the matrix $J_k^3(0)$.

5. Let $A \in M_n$ be nilpotent, so $A^k = 0$ for some k. Show that the characteristic polynomial of A is $p_A(t) = t^n$.

6. The linear transformation $d/dt: p(t) \to p'(t)$ acting on the vector space of all polynomials with degree at most 3 has the basis representation

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

in the basis $B = \{1, t, t^2, t^3\}$. What is the Jordan canonical form of this matrix?

7. What are the possible Jordan forms of a matrix $A \in M_n$ such that $A^3 = I$?

8. What are the possible Jordan canonical forms for a matrix $A \in M_6$ with characteristic polynomial $p_A(t) = (t+3)^4(t-4)^2$?

9. Use the method described in (3.2.1) to determine the Jordan canonical form of

$$A = \begin{bmatrix} i & 1 \\ 1 & -i \end{bmatrix}$$

10. Verify the assertion in the proof of Theorem (3.2.4.2) that products of matrices of the form (3.2.4.3) have the same form. Deduce that the inverse of a nonsingular matrix of the form (3.2.4.3) has the same form. *Hint:* The inverse of A is a polynomial in A.

11. Let $A, B \in M_n$. Use the identities in the proof of Theorem (1.3.20) to show that the nonsingular blocks in the Jordan forms of AB and BA are identical. Does this mean that AB and BA are similar? If AB and BA are not similar, discuss how far from similar they can be.

12. Suppose that $A_1, \ldots, A_k$ are given matrices with $A_i \in M_{n_i}$ for $i = 1, 2, \ldots, k$, and suppose that $J_1, \ldots, J_k$ are their respective Jordan canonical forms. Show that the Jordan canonical form of the direct sum

$$\begin{bmatrix} A_1 & & & 0 \\ & A_2 & & \\ & & \ddots & \\ 0 & & & A_k \end{bmatrix} \in M_n, \qquad n_1 + n_2 + \cdots + n_k = n$$

has (up to permutation of diagonal subblocks) the Jordan canonical form

$$\begin{bmatrix} J_1 & & & 0 \\ & J_2 & & \\ & & \ddots & \\ 0 & & & J_k \end{bmatrix}$$

13. Let $A \in M_n$ and $B, C \in M_m$ be given. Show that the direct sum $\left[\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right] \in M_{n+m}$ is similar to the direct sum $\left[\begin{smallmatrix} A & 0 \\ 0 & C \end{smallmatrix}\right]$ if and only if B is similar to C.

14. Let $B, C \in M_m$ be given. Show that the two k-fold direct sums

$$\begin{bmatrix} B & & & 0 \\ & B & & \\ & & \ddots & \\ 0 & & & B \end{bmatrix} \in M_{km} \quad \text{and} \quad \begin{bmatrix} C & & & 0 \\ & C & & \\ & & \ddots & \\ 0 & & & C \end{bmatrix} \in M_{km}, \qquad k \geq 1$$

are similar if and only if B and C are similar.

15. Let $A \in M_n$ and $B, C \in M_m$ be given. Show that the direct sums

$$\begin{bmatrix} A & & & 0 \\ & B & & \\ & & \ddots & \\ 0 & & & B \end{bmatrix} \in M_{n+km} \quad \text{and} \quad \begin{bmatrix} A & & & 0 \\ & C & & \\ & & \ddots & \\ 0 & & & C \end{bmatrix} \in M_{n+km}, \qquad k \geq 1$$

are similar if and only if B and C are similar.

16. Let $A \in M_n$ have Jordan canonical form $J_{n_1}(\lambda_1) \oplus \cdots \oplus J_{n_k}(\lambda_k)$. If A is nonsingular, show that the Jordan canonical form of A^2 is $J_{n_1}(\lambda_1^2) \oplus \cdots \oplus J_{n_k}(\lambda_k^2)$; that is, the Jordan canonical form of A^2 is composed of precisely the same collection of Jordan blocks as A, but the respective eigenvalues are squared. Is a statement like this true for all powers A^m, $m \geq 2$? Give a 2-by-2 example to show this is false if A is singular. *Hint:* If $\lambda \neq 0$, show that the Jordan canonical form of $J_k^2(\lambda)$ (a simple Jordan block) is $J_k(\lambda^2)$. Show that

$$\operatorname{rank}[J_k(\lambda) - \lambda I]^m = \operatorname{rank}[J_k^2(\lambda) - \lambda^2 I]^m, \qquad m = 1, 2, \ldots, k, \quad \text{if} \quad \lambda \neq 0$$

17. If $A \in M_n$, show that $\operatorname{rank} A = \operatorname{rank} A^2$ if and only if the geometric and algebraic multiplicities of the eigenvalue $\lambda = 0$ are equal; that is, all

the Jordan blocks corresponding to $\lambda = 0$ (if any) in the Jordan canonical form of A are 1-by-1.

Further Readings. For a proof of the fact that the similarity matrix between a given matrix and its transpose may always be taken to be symmetric, see O. Taussky and H. Zassenhaus, "On the Similarity Transformation between a Matrix and Its Transpose," *Pacific J. Math.* 9 (1959), 893–896. See [HJ] for a proof of the converse to Theorem (3.2.4.2) mentioned at the end of Section (3.2.4).

3.3 Polynomials and matrices: the minimal polynomial

If $p(t) = t^k + a_{k-1}t^{k-1} + a_{k-2}t^{k-2} + \cdots + a_1 t + a_0$ is a given polynomial, then one can always define

$$p(A) \equiv A^k + a_{k-1}A^{k-1} + a_{k-2}A^{k-2} + \cdots + a_1 A + a_0 I$$

for any $A \in M_n$. There is an important interplay between polynomials and matrices. The vital role of the characteristic polynomial has already been observed, but there are other polynomials associated with a square matrix. One of these is the minimal polynomial.

The Cayley–Hamilton theorem (2.4.2) guarantees that for each $A \in M_n$ there is a polynomial (the characteristic polynomial) $p_A(t)$ of degree n such that $p_A(A) = 0$. A polynomial whose value is the 0 matrix at A is said to *annihilate* A. There may also be a polynomial of degree $n-1$ which annihilates A, or one of degree $n-2$, but it is clear that, since there are only finitely many possibilities, for each $A \in M_n$ there is a polynomial of minimum degree that annihilates A, and this minimum degree is at most n. If $p(A) = 0$, then $cp(A) = 0$ for any $c \in \mathbf{C}$, so it is clear that we may always normalize a nontrivial annihilating polynomial so that the coefficient of the highest-order term is $+1$. A polynomial whose highest-order term has coefficient $+1$ is said to be *monic*. Notice that a monic polynomial cannot be identically zero.

3.3.1 **Theorem.** Let $A \in M_n$ be given. There exists a unique monic polynomial $q_A(t)$ of minimum degree that annihilates A. The degree of this polynomial is at most n. If $p(t)$ is any polynomial such that $p(A) = 0$, then $q_A(t)$ divides $p(t)$.

Proof: The characteristic polynomial is an example of a polynomial of degree n that annihilates A, so there is a minimum positive integer $m \le n$ such that there exists a monic polynomial $q(t)$ of degree m with $q(A) = 0$.

If $p(t)$ annihilates A, and if $q(t)$ is a monic polynomial of minimum degree that annihilates A, then the degree of $q(t)$ must be less than or equal to the degree of $p(t)$. By the Euclidean algorithm, therefore, there exists a polynomial $h(t)$ and a polynomial $r(t)$ of degree less than that of $q(t)$ such that $p(t) = q(t)h(t) + r(t)$. But $0 = p(A) = q(A)h(A) + r(A) = 0h(A) + r(A)$, so $r(A) = 0$. If $r(t) \not\equiv 0$, we could normalize it and obtain a monic polynomial of degree less than that of $q(t)$ that annihilates A. Since this would contradict the minimal property of $q(t)$, we conclude that $r(t) \equiv 0$, and hence that $q(t)$ divides $p(t)$ with quotient $h(t)$. If there are two monic polynomials of minimum degree that annihilate A, this argument shows that each divides the other; since the degrees are the same, one must be a scalar multiple of the other. But since both are monic, the scalar factor must be $+1$ and they are identical. □

3.3.2 **Definition.** Let $A \in M_n$ be given. The unique monic polynomial $q_A(t)$ of minimum degree that annihilates A is called the *minimal polynomial* of A.

3.3.3 **Corollary.** Similar matrices have the same minimal polynomial.

Proof: If $A, B, S \in M_n$ and if $A = SBS^{-1}$, then $q_B(A) = q_B(SBS^{-1}) = Sq_B(B)S^{-1} = 0$, so the degree of $q_B(t)$ is not less than the degree of $q_A(t)$. But $B = S^{-1}AS$, so the same argument shows that the degree of $q_A(t)$ is not less than the degree of $q_B(t)$. Thus, these two monic polynomials have the same minimal degree and both annihilate A, so they must be identical by Theorem (3.3.1). □

3.3.4 **Corollary.** For every $A \in M_n$, the minimal polynomial $q_A(t)$ divides the characteristic polynomial $p_A(t)$. Moreover, $q_A(\lambda) = 0$ if and only if λ is an eigenvalue of A, so every root of $p_A(t) = 0$ is a root of $q_A(t) = 0$.

Proof: Since $p_A(A) = 0$, the fact that there is a polynomial $h(t)$ such that $p_A(t) = h(t)q_A(t)$ follows from the theorem. This factorization makes it clear that every root of $q_A(t) = 0$ is a root of $p_A(t) = 0$, and hence every root of $q_A(t) = 0$ is an eigenvalue of A. If λ is an eigenvalue of A, and if $x \neq 0$ is an associated eigenvector, then $Ax = \lambda x$ and $0 = q_A(A)x = q_A(\lambda)x$, so $q_A(\lambda) = 0$. □

This last corollary shows that if the characteristic polynomial $p_A(t)$ has been completely factored as